

Decentralization Optimization Model Results – Colombia

We have run the decentralization optimization model on 14 Dec 2023 with the following parameters:

```
#def main(file_path_current_nodes):
    """Main function to execute the analysis and visualization pipeline."""
    # Step 1: Input data
    df_nodes = pd.read_csv('current_nodes_20231214_145236.csv')

    # Load pipeline of nodes which is not yet in the registry
    df_node_pipeline = get_node_pipeline()

    # Define candidate nodes which you would like to add
    df_candidate_nodes = create_candidate_node_dataframe(node_provider='Geeta
    Kalwani',
                                                         data_center='bg1',
                                                         data_center_provider='EdgeUno',
                                                         country='CO',
                                                         is_sev = True,
                                                         no_nodes = 4)

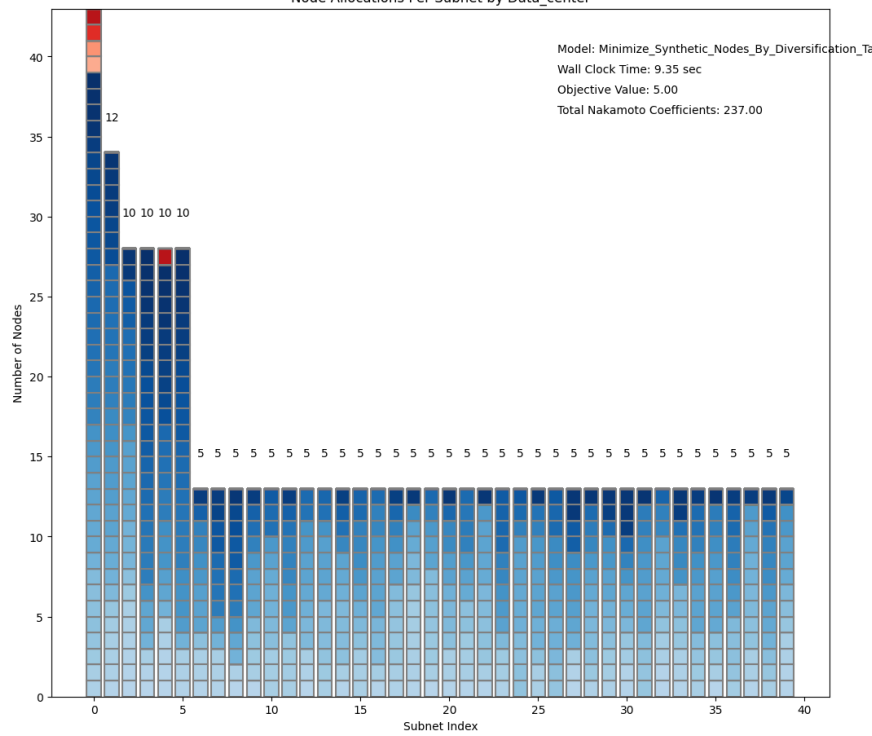
    # Set target topology
    subnet_limits = {
        "node_provider": 1,
        "data_center": 1,
        "data_center_provider": 1,
        "country": 2
    }
    network_topology = get_target_topology(subnet_limits)
    network_topology.loc[network_topology['subnet_size'] >= 28, 'subnet_limit_country']
    = 3 = 3
```

ObjectiveValue prior to running the model with the above parameters: 9

ObjectiveValue after running the model with the above parameters: 5

Result - Optimal solution found

Objective value:	5.00000000
Enumerated nodes:	0
Total iterations:	0
Time (CPU seconds):	4.23
Time (Wallclock seconds):	4.41



	Data_centers			
mn2	zh4	wa2	DC23	
j02	hu1	pa2	DC24	
bt1	zh5	wy1	DC25	
at2	bn1	li1	DC26	
nm1	zh2	ta2	DC27	
zh3	ch2	pa1	DC28	
hk1	dl1	ma2	DC29	
ge1	st1	"gn1"	DC30	
an1	bo1	"kr1"	DC31	
mm1	fr2	"cm1"	DC32	
bc1	ny1	pl2	DC33	
br1	sh1	cr1	DC34	
to2	sg3	tv1	DC35	
ty2	pr1	me3	DC36	
y1	hk3	pc1	DC37	
zh6	nd1	bg1	DC38	
jl1	sj3	DC1	DC39	
sj1	ty3	DC2	DC40	
or1	jb3	DC3	DC41	
ph1	mb1	DC4	DC42	
mr1	lj2	DC5	DC43	
sg1	br2	DC6	DC44	
ch3	ba1	DC7	DC45	
pl1	tb1	DC8	DC46	
hk4	zh7	DC9	DC47	
to1	rg1	DC10	DC48	
ch2	mt1	DC11	DC49	
ty1	gn1	DC12	DC50	
at1	lv1	DC13	DC51	
ge2	sg2	DC14	DC52	
jb1	ny2	DC15	DC53	
fm1	kr1	DC16	DC54	
mu1	ct1	DC17	DC55	
bu1	cm1	DC18	DC56	
sj2	ma1	DC19	DC57	
vl2	sl1	DC20	DC58	
aw1	ta1	DC21	DC59	
sc1	rg3	DC22	DC60	
tp1				

